Seminar: Deep Learning for Molecular Biology

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Helmholtz Center for Infection Research & TU Braunschweig
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Takeaways

Basic Knowledge of Machine Learning: Understanding of fundamental concepts in machine learning.

- A hot topic in both scientific research and industry applications.
- Wide-ranging impact across various domains, from healthcare to finance.

Basic Knowledge of Deep Learning: Familiarity with the principles of deep learning.

- Specialized models built upon neural network architectures.
- Remarkable success in tackling complex and challenging problems across domains.

Basic Knowledge of Bioinformatics

- Understanding of bioinformatics principles and applications.
- Awareness of how machine learning and deep learning are applied in bioinformatics research.

Experience in Problem Solving:

- Practical experience in applying machine learning and deep learning techniques to bioinformatics problems.

Seminar overview

Max. number of participants: 10

Language: English

Designated for Bachelor and Master students of Computer Science and Data Science

Deliverables:

- Joint presentation of the chosen topic (50% by each student, on seminar day)
  - covering practical (implementation in Python) and theoretical parts (split topics).
  - Presentation of 40 minutes: 20 mins theory, 20 mins results plus showing the codes

Seminar overview

- 7-10 page scientific report
  - written jointly, 50% by each student
  - Introduction, methods, results, discussion, literature (scientific references), acknowledgements (in report and table)
  - Introduction should cover both biology (superficially) and deep learning theory
  - Acknowledgement of the used tools (which and how, e.g. ChatGPT), you should take full ownership of the report, it means understand and approve every sentence there and know every reference you cite.
- Attend midterm meeting with the assistants
- Due Dates
  - First report version: 1 week before midterm meeting
  - Final report version: 2 weeks before the seminar date
- Grade= 50% presentation, 50% report

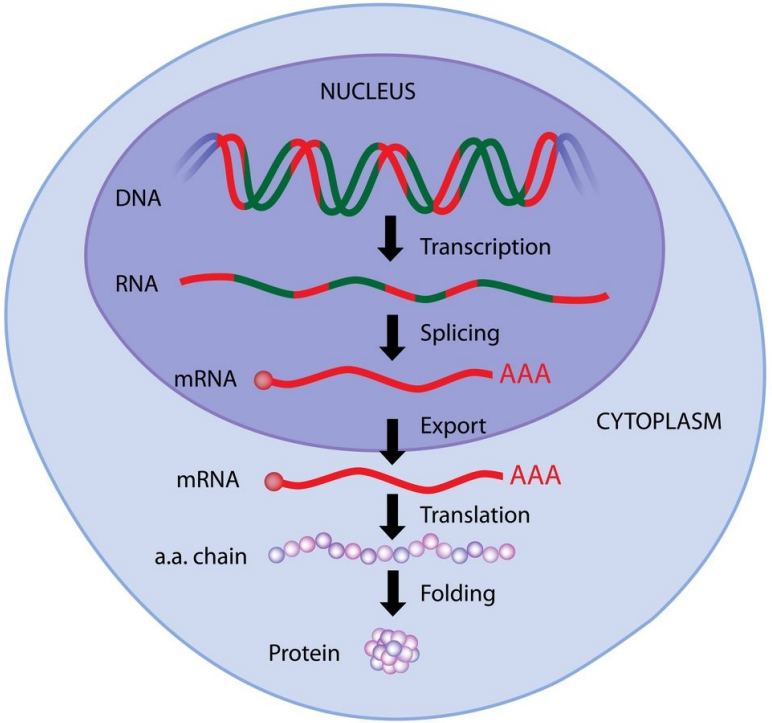
Introduction to Bioinformatics, Machine Learning, and Deep Learning

What is bioinformatics

(Current) Biology =

- DNA + RNA + Protein + Interactions

Bioinformatics: Computational analysis of the biological data



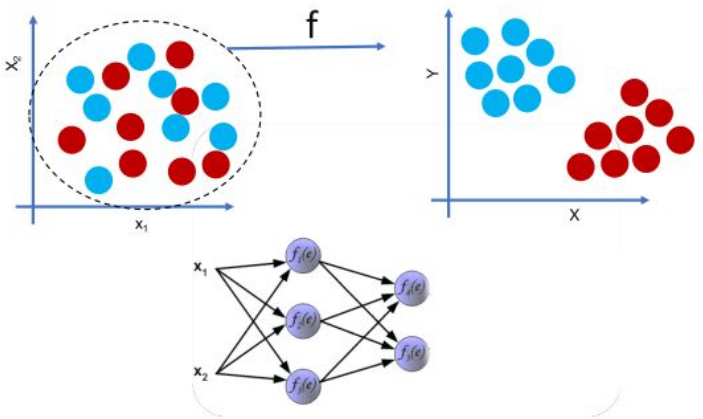
Machine Learning & Representation learning

Machine Learning: Learn patterns, representations, or behaviors from empirical data

- Constructs models that generalize from training data to unseen instances
- Includes supervised, unsupervised, and reinforcement learning

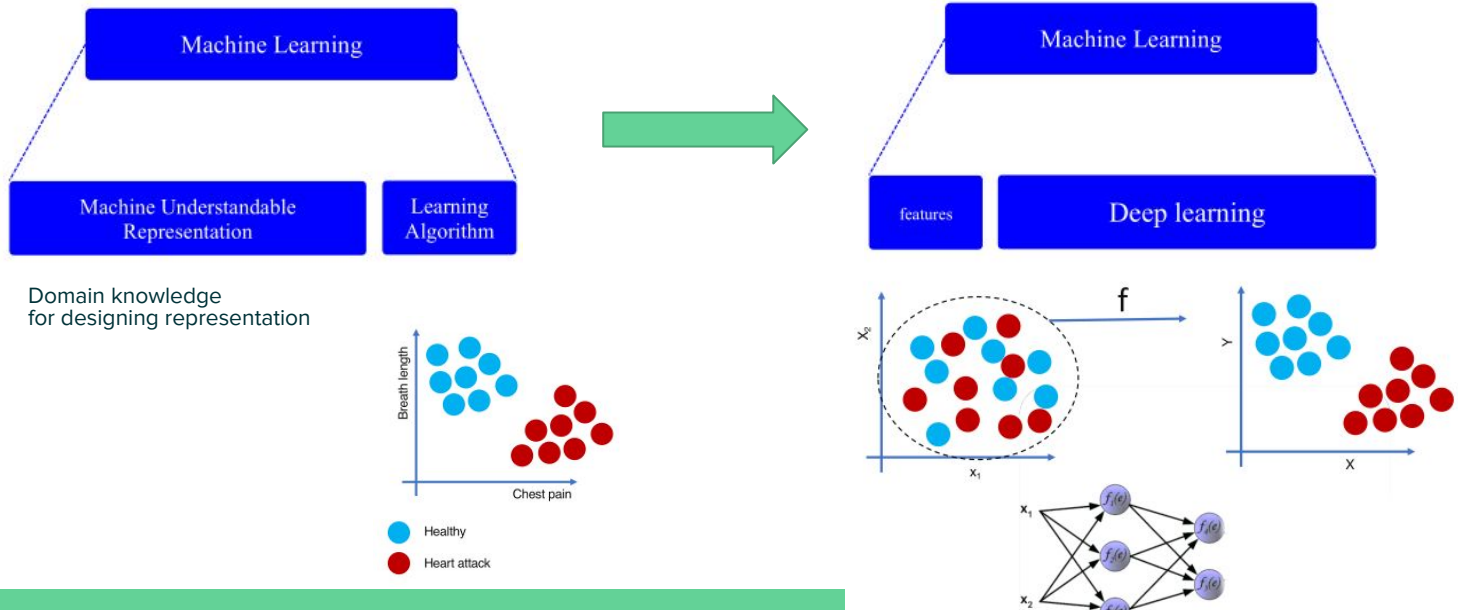
Representation learning:

- Data can be represented as points in n-dimensional space (of features)
- Learns hierarchical or latent representations that capture underlying structures in data
    - Then, we can solve several problems, e.g. classification.



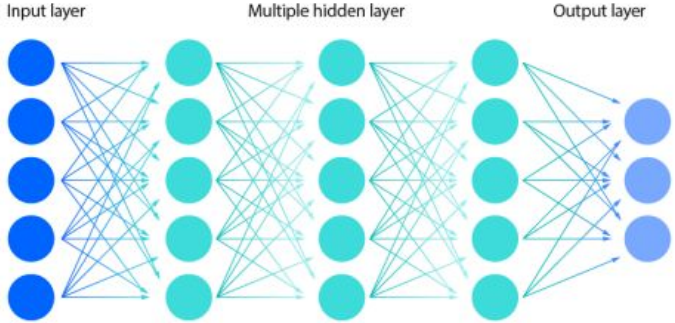
What is machine learning

- We alter data appearance to be interpretable by the audience.
  - Machine as the audience? Numerical values, vectors, matrices
- Finding a proper representation has been critical in machine learning

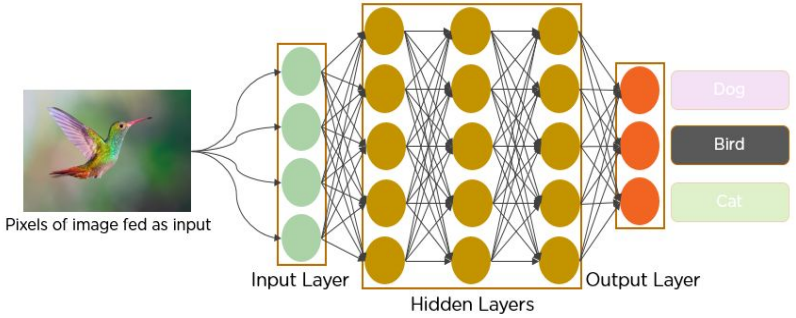


Specific ML models: NN, RNN, CNN

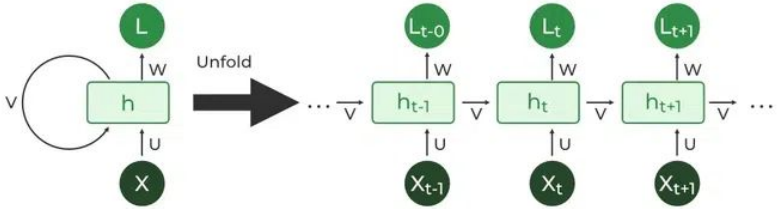
Neural network (NN): Inspired by brain



Convolutional Neural Network (CNN): NN with fewer (redundant) parameters

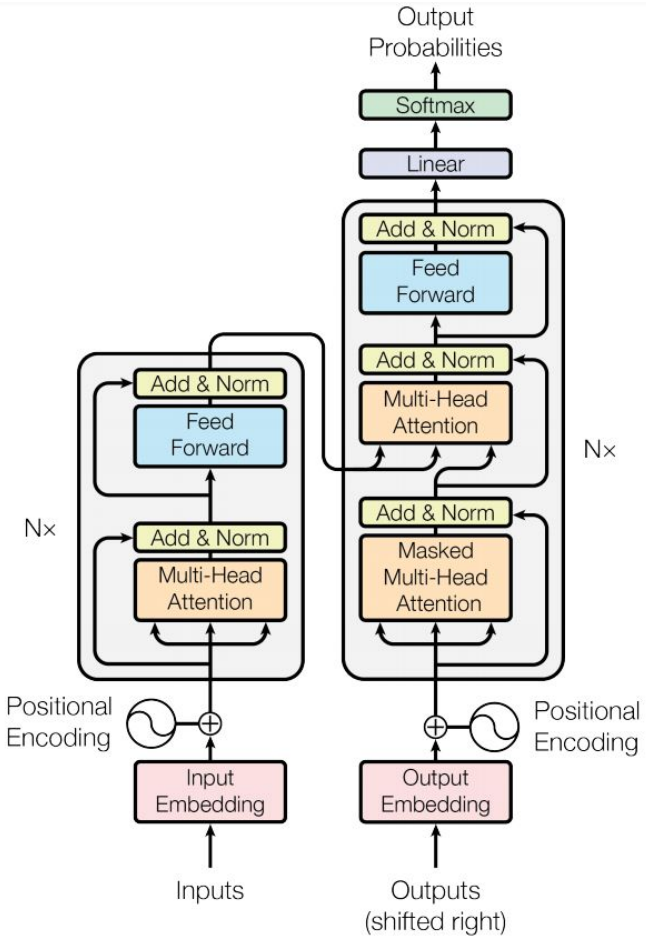


Recurrent Neural Network (RNN): Dealing with sequential information



What is a transformer?

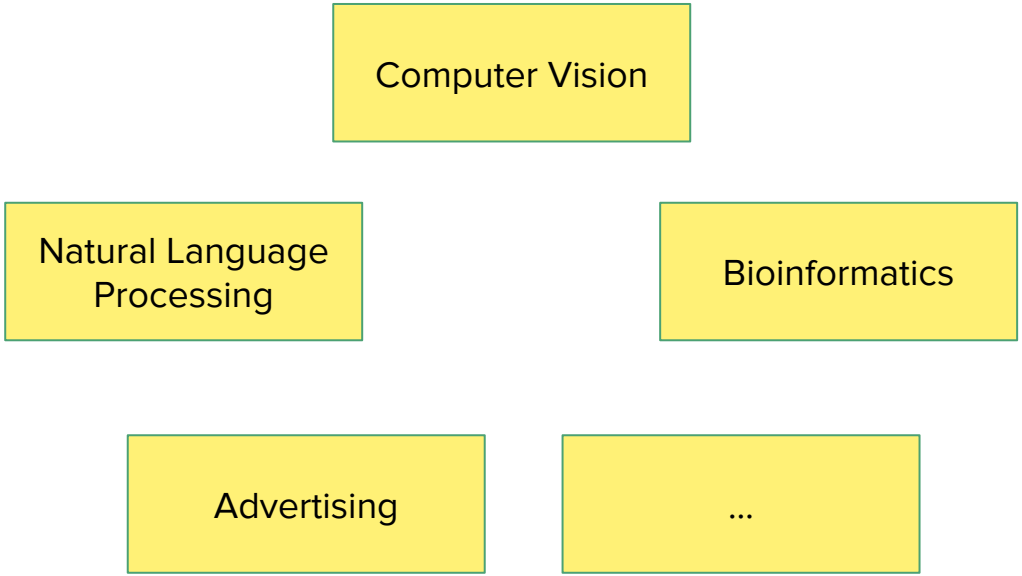
- A specific architecture with breakthrough performances
- Encoder/Decoder architecture
- Used in a wide range of applications
- Effective
- Highly parallelizable
- Ideal for transfer learning via representation learning



Transformers model architecture.

Vaswani et al, Attention is all you need. (NIPS 2017)

Applications of deep learning



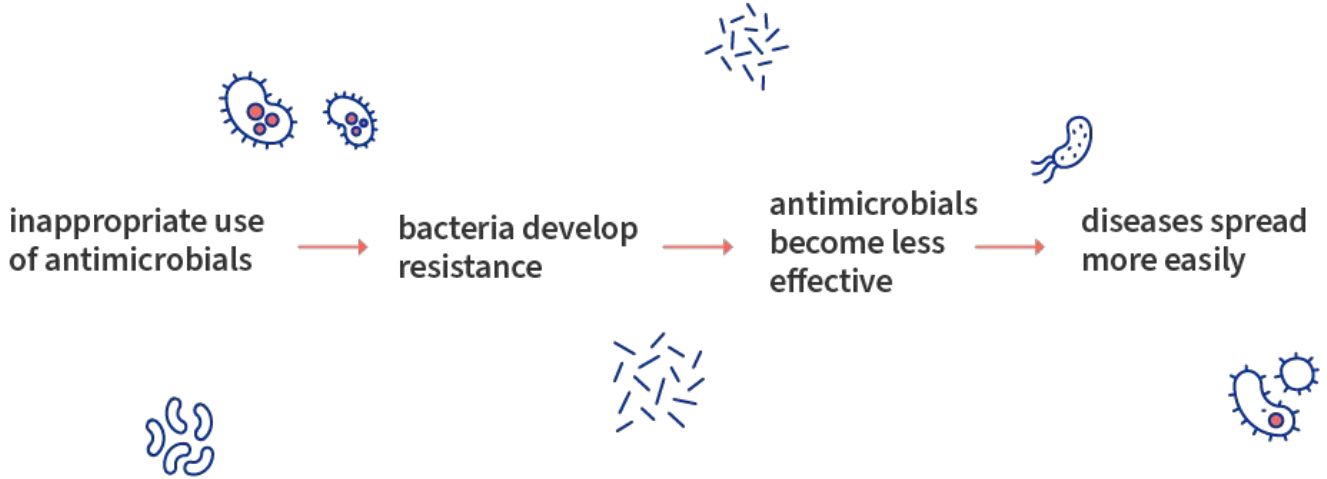
Your task

- Find a partner to form a group
- Choose a topic (from provided list)
- Study the topic
  - Some literature is provided
  - Meet and consult with the lecturers (at least once)
  - Implement and test your topic with the dataset (if applicable)
  - Evaluate (Metrics, precision/recall, TP, TN, ...)
- Create a written report
  - **Using AI tools is permitted; please detail their use in the acknowledgements section!**
- Present your topic on the presentation day

Implementation Tasks

Antimicrobial Resistance of pathogens (AMR)

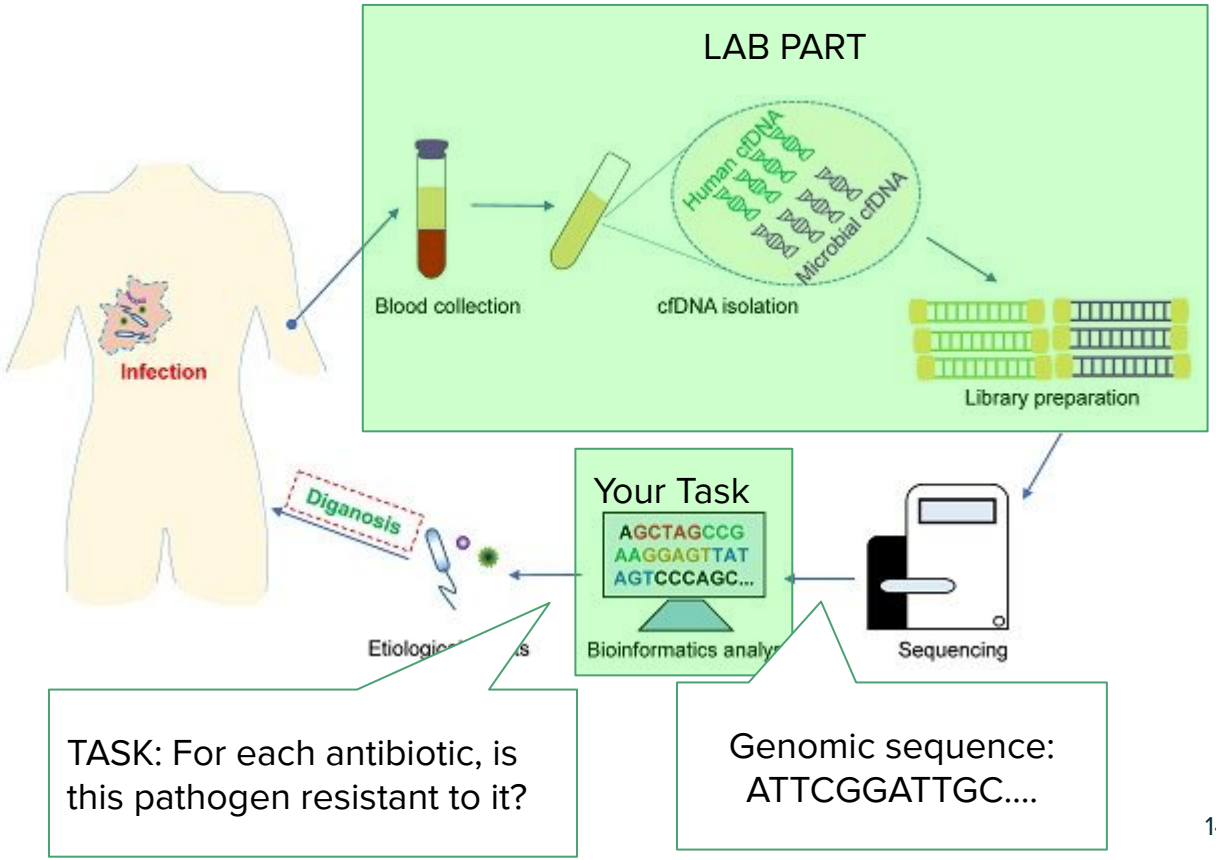
> “AMR occurs when bacteria, viruses, fungi and parasites change over time and no longer respond to medicines making infections harder to treat and increasing the risk of disease spread, severe illness and death.” WHO



Implementation Tasks

Antimicrobial Resistance of pathogens (AMR):

- Input: pathogen information
- Output: Is it resistance to specific antibiotics?



Source: https://www.thno.org/v10p5501.htm

Datasets description (First dataset)

- Git repository: https://github.com/hzi-bifo/seminar-dlmb-2026-winter-public
- `Staphylococcus_aureus_cefoxitin_pbp4` dataset
  - Predict AMR for a pathogen (*Staphylococcus aureus*) against an antibiotic (Cefoxitin) given the genomic sequence of one gene (pbp4)
  - Training set (800 samples) and test set (99 samples)
  - Labels: 0 (non-resistant), 1 (resistant)
- This dataset is the mandatory dataset.

Datasets description (Second dataset)

- `Klebsiella_pneumoniae_aztreonam_gene_bla` dataset
  - Predict AMR for a pathogen (*Klebsiella_pneumoniae*) against an antibiotic (Aztreonam) given the genomic sequence of a few of its genes as input
    - bla_1, bla_2, bla_3, ampC, ampD, ampE, ampG, ampR, acrB, acrA, acrR, baeR, baeS, cpxA, cpxR, ramA, cpxA, cpxR
  - Training set (1136 samples) and test set (142 samples)
  - Labels: 0 (non-resistant), 1 (resistant)
- This dataset is a mandatory dataset.

Datasets description (Optional dataset)

- `Klebsiella_pneumoniae_aztreonam` dataset
  - Predict AMR for a pathogen (*Klebsiella pneumoniae*) against an antibiotic (Aztreonam) given the genomic sequence of all its genes as input
    - Note that each pathogen's genome is given as a set of contigs
  - Training set (1136 samples) and test set (142 samples)
  - Labels: 0 (non-resistant), 1 (resistant)
- This dataset is an optional dataset.

Topics

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- Feed-forward Neural Networks and back propagation [2 students]
- Convolutional Neural Networks (CNNs) [2 students]
- Recurrent Neural Networks and LSTMs [2 students]
- Transformers - Encoder [2 students]
- Transformers - Decoder, Encoder-Decoder [2 students]

Feed-forward Neural Networks and back propagation [2 students]

Goals: Getting familiar with the basics of neural networks

- Introduction to linear classification, multilayer perceptron (MLP), and back propagation algorithm
- Implementation of MLP in python

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Suggested references

- Linear prediction: Lec. 1,2 at (https://bit.ly/1DIpc51) and (Chapter 4: https://stanford.io/2voWjra)
- G. Hinton's lecture 2, 3: https://bit.ly/3TNBPqw
- Lecture from U of Waterloo: https://bit.ly/2A2mzgN
- Stanford Tutorial: https://stanford.io/1FRrkZw
- Practicals: In Keras (keras.io) or Pytorch (pytorch.org)

Convolutional Neural Networks (CNN) [2 students]

Goals: Getting familiar with the convolutional neural network

- CNN architecture
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- G. Hinton's lecture: https://bit.ly/3TNB
- Stanford Tutorial: https://stanford.io/1FRrkZw
- A more advanced reference: deeplearningbook.org
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Recurrent Neural Networks [2 students]

Goals: Getting familiar with the RNNs and in particular LSTM

- Understanding “Vanilla” RNN

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- Lecture from U of Waterloo: https://bit.ly/2RCNEhn
- Understanding LSTMs: https://bit.ly/1S6gmjZ
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Suggested references

- Representation learning: https://arxiv.org/pdf/1206.5538.pdf
- Transformer paper: https://arxiv.org/abs/1706.03762
- Simple explanation of transformers: https://jalammar.github.io/illustrated-transformer/
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Goals: Getting familiar with the concept of these architectures

- Decoder-only models (GPT)
- Encoder-decoder models
- Their applications in bioinformatics, including state of the art articles

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Further steps

Send an email (to both of us)
- From one member CC all other members
- 2 preferred topics
- Until **24.10**

Split pairs: theoretical vs. practical tasks

Meeting and consultation with the lecturers (at least once)
Let us know in case of conflicts within the team

Send **report and codes** two weeks before the seminar day

Final seminar date:
- TBD 9:00 - 11:00 at BRICS (30.01 to 04.02).

Any questions? Contact us:
- mohammad-hadi.foroughmand-araabi@helmholtz-hzi.de
- georgios.kallergis@helmholtz-hzi.de

Thank you for your attention
Good luck!

References

- Deep learning in Life Science Youtube series
- AMR image https://www.consilium.europa.eu/en/infographics/antimicrobial-resistance/